

TRENDS IN THE PREVALENCE AND INCIDENCE OF DIABETES: NON-INSULIN-DEPENDENT DIABETES MELLITUS

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In contrast to insulin-dependent diabetes (IDDM), non-insulin-dependent diabetes (NIDDM) generally has its onset in later life. The two forms of diabetes are now seen as distinct disorders, both etiologically and pathologically, and this distinction has been of considerable importance in understanding their respective epidemiologies.

Whilst IDDM is rare in developing countries, NIDDM reaches its highest prevalence in non-European communities. Genetic predisposition to NIDDM appears to vary between populations, and in susceptible subjects certain potentially modifiable precipitants of the disease are strongly suspected. Of these, the three most important appear to be obesity, low levels of habitual physical activity, and aspects of the modern diet. The close association between these factors and the imprecision of their measurement have hindered the exploration of their influence in different populations.

Although there has been considerable research activity in the field of the epidemiology of NIDDM in the present decade, there has been a regional imbalance in this regard, and our knowledge of the subject remains very inadequate for large areas of the world.

Epidemiological considerations

A serious constraint upon the interpretation of reports of the frequency of diabetes is that criteria for the diagnosis of the disease have varied widely in the past. Many reports failed to record their criteria clearly, and many also failed to distinguish between NIDDM and IDDM.

For diabetes epidemiology, therefore, one of the major achievements of the present decade has been the provision of clear and unambiguous guidelines for the classification and diagnosis of the disease (1). Since it is well known that considerable discrepancies can occur in estimates of disease frequency when study methods and diagnostic criteria vary, this review will concentrate on some of the recent studies which have attempted to use the criteria recommended by WHO (1). For convenience, migrant communities, including the large non-indigenous populations in North America, Australia and New Zealand, will be considered together with other populations in their region of origin.

Most of the evidence regarding the frequency of NIDDM in populations comes from surveys carried out at a single point in time. This approach may be used to determine

NIDDM *prevalence* (i.e. the proportion of the population suffering from the disease at that point in time). The *incidence* of NIDDM (i.e. the number of new cases occurring in the population in a specified period of time) can only be calculated by monitoring the population continuously, or by repeated cross-sectional surveys. The resources required for such studies are often considerable. Furthermore, it is well known that a substantial proportion of subjects with NIDDM remain undetected in the community, and incidence estimates based upon routine data sources are much less reliable than they may be for IDDM.

For these reasons, comparatively few estimates of incidence of NIDDM have been reported. In this review we will therefore consider prevalence estimates for NIDDM first, before discussing the few incidence estimates that are available. Unfortunately, prevalence estimates also present problems in interpretation, even when standardized study methods have been used, since prevalence depends on both incidence and survival, and the latter can vary between populations for many reasons.

Since diabetes becomes more prevalent with advancing age in almost all communities, differences in age structure between populations can also lead to misleading comparisons, unless these differences are allowed for statistically. Although this has been done in some cases (in a series of standardized surveys in the various communities of the Western Pacific, for example), world-standardized rates are not yet available for NIDDM, and most estimates reported here are crude figures.

To date, few reports of incidence or prevalence of NIDDM have included confidence intervals or other estimates of precision. This creates further difficulties in the interpretation of population comparisons.

It is important to be aware of all these constraints when reviewing global patterns of a disease such as NIDDM.

Prevalence of NIDDM

Some recent estimates of the prevalence of NIDDM,^c using the WHO criteria, are summarized in Table 1.

Africa

Population-based reports of diabetes in Africa are uncommon, though a few have appeared recently.

Ahren & Corrigan (2) determined the prevalence of diabetes according to WHO criteria in the north-western part of the United Republic of Tanzania. Prevalence in subjects over the age of 20 was 1.6%. Three separate communities were included in the study, two of which were in rural locations. Interestingly, prevalence was 2.5% in the rural-dwelling Haya tribe, as compared with 0.5% in the rural Sukuma. Prevalence in the urban area of

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^c Prevalence of IDDM appears to be so low in most non-industrialized societies that prevalence of diabetes may be taken as an approximation to that of NIDDM in such situations.

TABLE 1. SOME RECENT ESTIMATES OF THE PREVALENCE (PER CENT POPULATION) OF NON-INSULIN-DEPENDENT DIABETES (NIDDM) IN VARIOUS POPULATIONS USING CRITERIA RECOMMENDED BY WHO^a

TABLEAU 1. ESTIMATIONS RÉCENTES DE LA PRÉVALENCE (POURCENTAGE DE LA POPULATION) DU DIABÈTE SUCRÉ NON INSULINO-DÉPENDANT (DSNID) DANS DIFFÉRENTES POPULATIONS, EN APPLIQUANT LES CRITÈRES RECOMMANDÉS PAR L'OMS^a

Region of origin/population Région d'origine/population	Country — Pays	Reference — Référence	Age range ^b Tranche d'âge ^b (years/ans)	M (%)	F (%)
Africa — Afrique					
Bantu — bantoue	United Rep. of Tanzania — République-Unie de Tanzanie	Ahren & Corrigan (2)	≥ 20	1.3	1.8
mixed (rural) — différentes ethnies (rurale)	Mali	Fisch et al. (3)	≥ 15	0.8	1.0
mixed (rural) — différentes ethnies (rurale)	Togo	Teuscher et al. (4)	all ages tous âges	0.0	0.0
black — noire	United States — Etats-Unis	Harris et al. (5)	20–74	8.6	11.0
Americas — Amériques					
Pima	United States — Etats-Unis	Knowler et al. (7)	≥ 20	32.6	37.2
Mapuche	Chile — Chili	Larenas et al. (9)	≥ 20	0.4	1.4
South-East Asia — Asie du Sud-Est					
Indian — indienne	Fiji — Fidji	Zimmet et al. (15)	≥ 20	12.7	12.9
rural — rurale				14.1	12.3
urban — urbaine					
Indian (urban) — indienne (ur- baine)	South Africa — Afrique du Sud	Omar et al. (16)	≥ 15	7.6	13.5
Indian — indienne	Singapore — Singapour	Thai et al. (19)	≥ 18	12.5	4.8
Malay — malaise	Singapore — Singapour	ibid.	≥ 18	9.1	6.1
Chinese — chinoise	Singapore — Singapour	ibid.	≥ 18	4.3	3.6
Indonesian (urban) — Indonésie (urbaine)	Indonesia — Indonésie	Waspadji et al. (20)	≥ 15	1.8	1.5
Europe					
rural — rurale	Italy — Italie	Verrillo et al. (12)	≥ 18	6.6	6.8
white — blanche	United States — Etats-Unis	Harris et al. (5)	20–74	5.5	7.3
white — blanche	Australia — Australie	Glatthaar et al. (14)	≥ 25	5.1	3.7
Eastern Mediterranean — Méditerranée orientale					
Arab — arabe	Tunisia — Tunisie	Papoz et al. (11)	≥ 20	2.3	0.6
rural — rurale				4.6	3.5
urban — urbaine					
Western Pacific — Pacifique occidental					
Micronesian — micronésienne	Nauru	Zimmet et al. (24)	≥ 20	24.6	23.9
Micronesian — micronésienne	Kiribati	King et al. (25)	≥ 20	3.6	3.6
rural — rurale				8.1	7.4
urban — urbaine				5.5	8.0
Polynesian — polynésienne . .	Rarotonga	King et al. (26)	≥ 20	5.3	9.1
	Niue	ibid.	≥ 20	1.4	3.5
	Wallis Island — Wallis	Taylor et al. (27)	≥ 20		
Melanesian (AN) ^c — mélané- sienne (AN) ^c	Fiji — Fidji	Zimmet et al. (15)	≥ 20	1.7	1.7
rural — rurale				4.8	8.2
urban — urbaine					
Melanesian (NAN) ^d — mélané- sienne (NAN) ^d	Papua New Guinea highlands — Papouasie-Nouvelle-Guinée, montagnes	King et al. (28)	≥ 20	0.0	0.0
Melanesian (NAN) — mélané- sienne (NAN)	Solomon Islands — Iles Salomon	Eason et al. (29)	≥ 18	0.0	1.4
rural — rurale				0.0	1.5
urban — urbaine					
Aboriginal (urban) — aborigène (urbaine)	Australia — Australie	Cameron et al. (30)	≥ 20	16.7	14.6

^a Reference (1) — Référence (1).

^b Note discrepancies in age ranges — Observer les différences de tranches d'âge.

^c AN: Austronesian ancestry — d'ascendance austronésienne.

^d NAN: non-Austronesian ancestry — d'ascendance non austronésienne.

Mwanza was 1.9%. Whilst the authors considered that these differences had implications for the planning of national health services, it appears from the report (2) that the differences between the three groups were not in fact statistically significant after allowing for differences in their age/sex distribution. This emphasizes the

need for careful evaluation of apparent differences in rates of disease. A further constraint in this study, which the authors acknowledged, was that fasting blood-glucose concentration was used as an initial screening test for glucose intolerance. This is known to be a relatively insensitive measure, as compared with the blood-

glucose concentration after an oral glucose challenge, and the prevalence of diabetes could therefore have been underestimated.

Another study which used the WHO criteria, but was based entirely upon fasting blood-glucose estimation, estimated prevalence of diabetes as 1% in rural Mali (3).

A further report from Western Africa, using WHO criteria and random blood-glucose estimation, failed to identify any diabetic subjects among 517 survey respondents in rural Togo (4). However, diabetes was known at the local hospital.

These three studies suggest that the prevalence of NIDDM is low in rural Africans at the present time. Although the difficulties attending field surveys in such areas should not be forgotten, these reports also illustrate some persisting discrepancies in study methods.

A report based upon the National Health and Nutrition Examination Survey (NHANES 2) suggests that NIDDM is more prevalent in black than in white North Americans in the United States, with approximately 10% of the former suffering from the disease (5). Thus, it may be that African genes carry greater underlying susceptibility to NIDDM than do Caucasian genes, and there may be a potential for a rise in the prevalence of NIDDM in Africa, as nations become more industrialized. There was the opposite suggestion, however, from the Mali data (3). In that survey, negroid ethnic groups appeared to be less affected than Caucasian or part-Caucasian communities. This raises the possibility that Caucasian genetic admixture might, in part, be responsible for the increased susceptibility of blacks in the United States.

A clinic-based report from Soweto (6) suggested a prevalence of known diabetes of only 0.1%. It also found approximately half of the diabetic subjects to be treated with insulin. Taken together, these two findings may suggest considerable under-ascertainment of milder cases of NIDDM in South African blacks.

Thus, the real status of NIDDM in the public health of Africa is far from clear. Further population surveys, particularly of urban communities and incorporating the recommended 75 g oral glucose challenge (1) could yield much useful information.

The Americas

There is abundant evidence that indigenous North Americans are highly susceptible to NIDDM. The community with the highest reported prevalence of NIDDM in the Americas is the Pima Indians of Arizona (7) in which approximately one-third of all adults are affected.

It also appears that for indigenous North Americans, susceptibility to NIDDM is related to proportional Amerindian genetic admixture, with subjects of mixed ethnicity having a risk of NIDDM intermediate between that of full-blood American Indians and Caucasians (8). There is some evidence to suggest that the prevalence of NIDDM is increasing in many North American Indian communities and that the Aleuts (related to the Eskimos) who were formerly thought to be relatively resistant to NIDDM, may now be suffering from it also.

The picture in South America is not so clear, and in many ways resembles the state of our knowledge of NIDDM in Africa. The prevalence of diabetes in the indigenous Mapuches of Chile was recently estimated as only 1% (9). A prevalence of 5% has been reported from La

Plata City (Argentina) (10) and this may have been an underestimate, since the survey was based upon a 50 g oral glucose challenge. Urban communities throughout Latin America would benefit from further investigation.

The Eastern Mediterranean

Little information is available on NIDDM in this Region also. A study in Tunisia, using WHO criteria, has recently estimated age-standardized prevalence as a percentage of the population to be 4.6 in males and 3.5 in females in urban Tunis, as compared with 2.3 in males and 0.6 in females in a rural area (11). Since these figures could be conservative, due to the use of fasting blood-glucose estimation only, NIDDM would appear to be of potential public health concern in urban populations in this Region also.

Europe, and populations of European origin

In Europeans, NIDDM is a relatively common disorder, but it does not reach the frequency seen in high-risk communities in the developing world. There have been surprisingly few population studies of NIDDM using WHO criteria in Europe itself.

A survey in rural Italy (12) estimated the prevalence of diabetes as 6.8% in females and 6.6% in males. A survey in Finland, a country with the highest reported prevalence of IDDM in the world, suggested that approximately one-third of elderly male Finns may have NIDDM (13). Recent information from Malta (A. Schranz, personal communication) suggests a prevalence of 10% in the total adult population of the island.

For whites in the United States, data on the prevalence of NIDDM were obtained nationwide during the NHANES 2 survey (5). This estimated the prevalence of diabetes in the 20-75 years age range as 6.6%, according to WHO criteria, which is very close to the Italian figure (12). Approximately one-half of subjects were undiagnosed prior to the survey. Prevalence rose from 2% in the 20-44 years age range, to 18% in the 65-74 years age range.

Thus, at the present time, almost one-fifth of white North Americans can expect to develop NIDDM if they live to the seventh decade of life. This somewhat alarming finding highlights the potential for an escalation in prevalence of NIDDM in European communities, as a result of the increasing longevity which most industrialized societies are experiencing at present. A somewhat lower estimate of the prevalence of NIDDM has been reported for white Australians (14).

South-East Asia

Present evidence suggests that there is a spectrum of underlying genetic susceptibility to NIDDM in Asian populations. Several surveys of migrants from the Indian subcontinent have suggested that Indians are the most susceptible. According to WHO criteria, prevalence has been reported as 12-14% in Indian migrants to Fiji (15). A high prevalence has also been reported in Indians living in South Africa (16) and Suriname (17).

A number of population studies of diabetes have been reported from South-East Asia itself, but study methods have varied widely. These reports have been reviewed elsewhere (18).

A recent study in Singapore (19) found prevalence to be lowest in Chinese, highest in Indians and intermediate in Malays. There was also a suggestion of an apparent secular increase in prevalence since a previous survey a decade earlier.

A survey in urban Jakarta (Indonesia) (20) found a prevalence of only 1.6%, slightly lower than the figure of 2.7% reported for migrant Indonesians in Suriname (17). As already noted for Indians, it is not uncommon for migrant communities to show a higher prevalence of NIDDM than those they leave behind, since migration is often accompanied by socioeconomic advancement.

As in the Singapore study, the conclusions of the majority of comparative studies in South-East Asia have been that Chinese have the lowest prevalence of diabetes in the Region, with other populations having rates intermediate between the high rates of the Indians and the low rates of the Chinese. However, recent reports from Taiwan (21) and Hong Kong (22) have recorded a prevalence of approximately 10% in urban Chinese over the age of 60 years, suggesting that Chinese may be more susceptible than was formerly supposed.

The Western Pacific

There have been a number of studies of diabetes in the Western Pacific using the WHO criteria, and our knowledge of the prevalence of NIDDM in this Region is becoming relatively complete. This large data base will be of great value in due course, for monitoring temporal trends and evaluating the effects of interventions aimed at primary and secondary prevention of the disease. Age-standardized prevalence according to WHO criteria in adults in 17 communities from 10 Pacific nations have been published elsewhere (23).

The highest prevalence of NIDDM in the Western Pacific has been reported from the Micronesian population of Nauru (24) where approximately one-quarter of all adults are affected. Other Micronesian populations are also very susceptible. In neighbouring Kiribati, approximately 8% of urban residents of both sexes were found to have NIDDM, as compared with 4% of residents of a more traditional island (25). When standardized for age, the risk for urban dwellers was threefold that of the rural community.

For Polynesia, a recent report estimated prevalence as approximately 5% in males and 8% in females in both Rarotonga (Cook Islands) and Niue (26). Taylor et al. (27) have demonstrated a low prevalence among traditional-living Polynesians on Wallis Island.

There is still some uncertainty as to the true susceptibility of Melanesians to NIDDM. Though the majority (of Austronesian ancestry) may be as susceptible as Polynesians (15), there is some evidence that the few Melanesian communities that lack Austronesian genetic ancestry (nonAustronesians) may be relatively resistant to NIDDM (28, 29).

NIDDM is now common in urbanized Australian Aborigines (30).

Incidence of NIDDM

There have been very few population-based estimates of NIDDM incidence, due to the practical difficulties of ascertaining occurrence of NIDDM in populations, when a substantial proportion of subjects meeting the WHO criteria for the disease remain asymptomatic. For the

United States, an estimate of physician-diagnosed diabetes—which may be expected to underestimate total incidence by as much as 50% (5)—gave an incidence of 117 per 100 000 person-years for three communities in Minnesota (31). This figure concurs with a previous report from Rochester, Minnesota.

Both the Pima Indians of Arizona and the Nauruans of Micronesia have been intensively studied with respect to NIDDM, and incidence estimates are available for each. For the Pimas (7) incidence was reported as 1.85 per 100 person-years at all ages, and for the Nauruans (32) 1.6 per 100 person-years in subjects aged 20 years and older (note the thousand-fold difference in the denominator, compared with the Minnesota estimate). From the data published (7) incidence in Pimas aged 25 and over could be calculated as 4.6 per 100 person-years. These high incidence estimates mirror the very high prevalence of NIDDM in these two unusual populations.

Since prevalence is the product of average incidence and average duration of disease, and considering their respective incidence and prevalence rates, average survival with established NIDDM may be calculated as approximately 15 years for the Nauruans, but only 8 years for the Pimas. The reason for the lower apparent life expectancy of Pima diabetics is not at all clear, and this should stimulate further comparisons between longitudinal studies of NIDDM in different populations. These may not only lead to further etiological insight, they may also provide greater understanding of important public health aspects of the disease.

Impaired glucose tolerance

The WHO classification of glucose intolerance (1) includes a category that is intermediate between normal tolerance and diabetes mellitus, called impaired glucose tolerance (IGT). Subjects in this range, whilst not considered to be suffering from diabetes, are known to be at substantially higher risk of developing the disease and its complications than are persons who have glucose tolerance in the normal range.

IGT has an unpredictable course, and is thought to represent a heterogeneous group of subjects, some of whom may be in progress towards frank diabetes, some of whom may have a transient abnormality and be destined to return to normality, and some of whom may continue in a steady IGT state.

Surveys which have used a 75 g oral glucose challenge and have estimated the two-hour blood-glucose value (but not those which have only studied the fasting, or random blood-glucose value) can assess the prevalence of IGT according to the WHO criteria. Several reports already discussed here with respect to NIDDM have also produced corresponding figures for IGT (Table 2).

None of the three African studies mentioned earlier included estimates of IGT. However, for the black population of the United States (5) NHANES 2 data indicate a prevalence of IGT of 13%, i.e. slightly higher than that of NIDDM. From the same study, prevalence of IGT in the white population was estimated as 11%, close to the figure for blacks, despite the whites' relatively lower prevalence of NIDDM.

The surveys of rural Italians (12) and white Australians (14) estimated prevalence of IGT as 6.4% and approximately 4% respectively.

In the Western Pacific, the prevalence of IGT in Fiji Indians (15) and Nauruans (24)—approximately 10%

TABLE 2. SOME RECENT ESTIMATES OF THE PREVALENCE (PER CENT POPULATION) OF IMPAIRED GLUCOSE TOLERANCE (IGT) IN VARIOUS POPULATIONS USING CRITERIA RECOMMENDED BY WHO^a

TABLEAU 2. ESTIMATIONS RÉCENTES DE LA PRÉVALENCE (POURCENTAGE DE LA POPULATION) DE L'ABAISSEMENT DE LA TOLÉRANCE AU GLUCOSE (ATG) DANS DIFFÉRENTES POPULATIONS, EN APPLIQUANT LES CRITÈRES RECOMMANDÉS PAR L'OMS^a

Population	Country — Pays	Reference — Référence	Age range ^b Tranche d'âge ^b (years/ans)	M (%)	F (%)
black — noire	United States — Etats-Unis	Harris et al. (5)	20-74	11.3	13.6
white — blanche	United States — Etats-Unis	Harris et al. (5)	20-74	10.2	11.1
rural — rurale	Italy — Italie	Verrillo et al. (12)	≥ 18	5.1	7.7
white — blanche	Australia — Australie	Glatthaar et al. (14)	≥ 25	4.3	3.3
Indian — indienne	Fiji — Fidji	Zimmet et al. (15)	≥ 20		
rural — rurale				10.4	10.8
urban — urbaine				9.4	11.2
Micronesian — micronésienne	Nauru	Zimmet et al. (24)	≥ 20	18.4	18.3
Micronesian — micronésienne	Kiribati	King et al. (25)	≥ 20		
rural — rurale				12.5	14.3
urban — urbaine				15.0	16.9
Polynesian — polynésienne	Rarotonga	King et al. (26)	≥ 20	9.7	10.0
Polynesian — polynésienne	Niue	ibid.	≥ 20	6.6	7.2
Polynesian — polynésienne	Wallis Island — Wallis	Taylor et al. (27)	≥ 20	4.0	9.3
Melanesian (AN) ^c — mélanésienne (AN) ^c	Fiji — Fidji	Zimmet et al. (15)	≥ 20		
rural — rurale				6.2	10.2
urban — urbaine				8.0	13.8
Melanesian (NAN) ^d — mélanésienne (NAN) ^d	Papua New Guinea highlands — Papouasie-Nouvelle-Guinée, montagnes	King et al. (28)	≥ 20	3.5	1.2
Melanesian (NAN) — mélanésienne (NAN)	Solomon Islands — Iles Salomon	Eason et al. (29)	≥ 18		
rural — rurale				0.2	0.7
urban — urbaine				0.0	0.0
Aboriginal, urban — aborigène, urbaine	Australia — Australie	Cameron et al. (30)	≥ 20	0.7	4.2

^a Reference (1) — Référence (1).

^b Note discrepancies in age ranges — Observer les différences dans les tranches d'âge.

^c AN: Austronesian ancestry — d'ascendance austronésienne.

^d NAN: non-Austronesian ancestry — d'ascendance non austronésienne.

and 18% respectively—reflects the high prevalence of NIDDM in these two populations.

Prevalence of IGT in several other Pacific populations of Micronesian, Polynesian and Melanesian ancestry (15, 25-30) varied widely, from almost total absence in the Solomon Islands (29) to 17% in urban females in Kiribati (25). IGT was present in the Papua New Guinea highlanders, although there was an absence of NIDDM (28). Compared with the high prevalence of NIDDM, IGT was sur-

prisingly uncommon in an urban Australian Aboriginal population (30).

Thus, there is a wide variation in the prevalence of IGT in populations, which to some extent mirrors their prevalence of NIDDM. However, this relationship is not entirely consistent, and the implications of the prevalence of IGT in the emergence of NIDDM in populations are intriguing, and not fully understood.

SUMMARY

The revision of the classification of diabetes mellitus, to differentiate clearly between insulin-dependent (IDDM) and non-insulin-dependent diabetes mellitus (NIDDM), and the provision of unambiguous guidelines for diagnosis (1) constitute important recent developments in diabetes epidemiology. However, our knowledge even of the prevalence of NIDDM remains surprisingly incomplete for many areas of the world.

Whilst NIDDM may still be uncommon in rural Africa, prevalence has been reported as approximately 10% in blacks in the United States, indicating a need for more information regarding the prevalence of NIDDM in urban Africa. There is also little information with regard to urban communities in Latin America.

The highest prevalence of NIDDM is found in certain indigenous North American and Western Pacific socie-

ties. In extreme cases approximately one-third of the adult population now suffers from the disease.

NIDDM is also common (prevalence approximately 5%) in Europe and in communities of European origin. Data from the United States suggest that approximately one-fifth of white North Americans can expect to develop NIDDM if they live to the seventh decade of life.

Of populations of South-East Asian ethnicity, Indians appear to be the most susceptible. Indian migrants to Fiji, South Africa and South America all demonstrate prevalence of NIDDM of 10% or more. Whilst it was formerly believed that Chinese were rarely affected by NIDDM, recent reports cast some doubt on this.

Prevalence of NIDDM in the Western Pacific varies widely. However, with the possible exception of certain Melanesian populations, prevalence is high in all com-

munities which have abandoned their traditional life-style.

Very few estimates of the incidence of NIDDM have been published, largely because of the practical difficulties attending the estimation of incidence in this often asymptomatic disease.

An intermediate category of impaired glucose tolerance

(IGT) has been recommended recently, to define subjects whose glucose tolerance, though outside the normal range, is not thought to be sufficiently deranged to warrant their being classified as diabetic. IGT appears to have an unpredictable course, though the risk of subsequent diabetes is substantially higher than for subjects with normal glucose tolerance. Reports indicate that the prevalence of IGT in different populations varies as widely as does that of NIDDM.

RÉSUMÉ

Tendances observées dans la prévalence et l'incidence du diabète: diabète sucré non insulino-dépendant

La révision de la classification du diabète sucré afin d'établir clairement la différence entre diabète sucré insulino-dépendant (DSID) et diabète sucré non insulino-dépendant (DSNID), et la formulation de directives sans ambiguïté pour le diagnostic (1) sont des faits notables récemment survenus dans l'épidémiologie du diabète. Toutefois, nos connaissances sur le DSNID dans de nombreuses parties du monde, s'agissant même de sa prévalence, restent étonnamment incomplètes.

Le DSNID est peut-être encore rare dans les régions rurales de l'Afrique, mais sa prévalence est évaluée à environ 10% chez les noirs des Etats-Unis d'Amérique, ce qui indique qu'il faudrait réunir davantage de données sur sa prévalence dans les zones urbaines de l'Afrique. De même, on possède peu de renseignements sur le DSNID dans les communautés urbaines en Amérique latine.

C'est dans certaines sociétés indigènes d'Amérique du Nord et du Pacifique occidental que l'on enregistre la plus forte prévalence du DSNID. Dans les cas extrêmes, environ un tiers de la population adulte est aujourd'hui atteinte de cette maladie.

Le DSNID est également courant (prévalence d'environ 5%) en Europe et dans les communautés d'origine européenne. Selon des données concernant les Etats-Unis, environ un cinquième des Américains de race blanche peuvent s'attendre à développer un DSNID à partir de 70 ans.

Parmi les populations de l'Asie du Sud-Est, les Indiens semblent être les plus vulnérables. Chez les migrants indiens à Fidji, en Afrique du Sud et en Amérique du Sud, on observe une prévalence du DSNID de 10% ou plus. Alors que l'on pensait que les Chinois étaient rarement atteints de DSNID, de récents rapports remettent en cause cette opinion.

La prévalence du DSNID dans le Pacifique occidental est extrêmement variable. Toutefois, à l'exception possible de certaines populations mélanésiennes, elle est élevée dans toutes les communautés qui ont abandonné leur mode de vie traditionnel.

Très peu d'estimations de l'incidence du DSNID ont été publiées du fait, essentiellement, des difficultés pratiques que soulève l'évaluation de l'incidence de cette maladie souvent asymptomatique.

Il a été recommandé récemment d'introduire la catégorie intermédiaire d'abaissement de la tolérance au glucose (ATG) pour définir les sujets dont la tolérance au glucose se situe en dehors de la fourchette normale, sans être suffisamment altérée pour qu'ils soient classés dans la catégorie des diabétiques. L'ATG semble évoluer de manière imprévisible, encore que le risque d'apparition d'un diabète soit nettement plus élevé que chez les sujets ayant une tolérance normale au glucose. Selon les rapports, la prévalence de l'ATG chez différentes populations est tout aussi variable que celle du DSNID.

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